#1

Consider $\mathcal{F}$ on x, we get

$$(\partial_t^2 - |\xi|^2)\hat{u}(\xi,t) = 0 \quad \hat{u}(\xi,t) = \hat{f}(\xi)$$

on $t > 0$.

Solve this system, we get

$$\hat{u}(\xi,t) = A(\xi)e^{t|\xi|} + B(\xi)e^{-t|\xi|}$$

to make the inversion hold, we eliminate $A(\xi)$ term and leave $\hat{u}(\xi,t) = \hat{f}(\xi)e^{-t|\xi|}$. As in remark 7.28 of suppnote, one will see

$$\mathcal{F}^{-1}(e^{-t|\xi|}) = \frac{2^{(n/2)}}{\sqrt{\pi}}\Gamma(\frac{n+1}{2})\frac{t}{(t^2+|x|^2)^{(n+1/2)}}$$

, this should be the Poisson kernel we want which follow from $\widehat{f * g} = \hat{f}\hat{g}$. So we are done.

#2

Define $u_\epsilon := u + \epsilon|x|^2$, then one have $\Delta u_\epsilon = 2n\epsilon > 0$, but if one have a maximum point inside interior, its Laplacian must smaller or equal to 0. Then one can conclude that its maximum must on its boundary for u_ϵ, take limit, one have

$$u(x) \leq \max_{\partial U} u$$

Other side inequality directly been derived as $\partial U \subset \bar{U}$.

#3

On $[0,T] \times \mathbb{R}^n$, for any $T > 0$, we let $0 < a < b < t_0 < T$, $x_0 \in B(x_0, R)$, and define such cylinder as in Thm 1.50. Recall we have

$$g(x,t) = \frac{1}{(4\pi(t_0-t))^{n/2}}e^{-R^2/4(t-t_0)}$$

we directly look for the boundary integral on its side (third term), by direct calculation,

$$\partial_\nu g = \frac{-(x-x_0)}{2(t_0-t)}g$$

$$\begin{aligned} |f\partial_\nu g - g\partial_\nu f| &\leq C(1+\frac{R}{2(t_0-t)})e^{\epsilon|x|^2}\frac{1}{(t_0-t)^{n/2}}e^{-R^2/4(t_0-t)} \\ &\leq C(1+\frac{R}{2(t_0-b)})e^{2\epsilon|x_0|^2}\frac{1}{(t_0-b)^{n/2}}e^{-(1/4(t_0-a)+2\epsilon)R^2} \end{aligned}$$

as $|x|^2 \leq 2|x_0|^2 + 2R^2$. If T small enough s,t, $t_0 - a < \frac{1}{8\epsilon}$, as $R \to \infty$, the value always goes to 0 for gaussian type speed against polynomial type. Multiple time length and surface area, this integral goes to 0.

Then use the same argument as in Thm 1.50 (both side take limit for the remaining integrals), we get above result on [0,T].

Then consider

$$v(x,t) = u(x,t+T)$$

its easy to find required condition hold, then we continues to doing infinitely many times, we are done.